

Blackouts Preserving Rectangle Identifiability on Grid Points

A partial solution to Problem 001 of Kagey’s Open Problem Collection

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Abstract

We study Problem 001 of Kagey’s Open Problem Collection: black out as many points of an $n \times m$ grid as possible so that every possible rectangle drawn with its corners on grid points remains uniquely identifiable from its visible corners. We reduce the problem to a minimum hitting set of a difference hypergraph, solve the two-row case completely (the Strip Theorem: maximum blackout $m + 1$ with exactly $m \cdot 2^{m-1}$ optima for $m \geq 3$), report cross-verified exact values for grids through 6×6 , prove the 5×5 maximum blackout unique up to symmetry, record one natural conjecture refuted by pre-registered computation, and state two live conjectures with supporting data.

1 Introduction

The problem is posed with some alterations chosen for clarity as the following: Suppose you, the player, are shown an $n \times m$ grid of vertices. Each vertex has two states, ‘white’ or ‘blackened out’. A vertex that is blackened out is unable to convey any information. The game master asks us to ‘black out’, or turn off as many of the vertices as we like. Once we choose the ‘blackout’, which is simply the set of all vertices that are blackened out, the game master will draw a rectangle that we cannot see but know the corners of given that the corners are on a white vertex. The rectangle’s sides are not necessarily parallel to the grid axes. We must be able to guess what the rectangle’s corners are from this information with complete confidence. The question posed by Kagey [1] is how many vertices can be blackened out while every rectangle the game master might draw remains distinctly identifiable.

The order of play is important here. Because the blackout is chosen by us before the rectangle is chosen, a blackout is valid only if *no* two rectangles are confusable under it. Writing $C(R)$ for the corner set of a rectangle and S for the blackout, what the player sees is $C(R) \setminus S$; the requirement is that this assignment be injective on the set of rectangles. Section 2 makes this precise, and the rest of the paper studies the resulting extremal quantity.

Our results are of four kinds. Proved: the difference-set reduction (Lemma 1), which recasts the problem as a minimum hitting set, and the Strip Theorem (Theorem 1), which settles the two-row case completely, giving maximum blackout $m + 1$ with exactly $m \cdot 2^{m-1}$ optima for $m \geq 3$. Computed and cross-verified: exact values and optimum counts for all grids through 6×6 (Table 1), together with the fact that the 5×5 maximum blackout is unique up to the symmetries of the square. Conjectured: two formulas supported by the computations (Conjectures 1 and 2). Refuted:

*Every published computation is cross-verified by two independently written implementations (the author’s ILP engine and an independently authored reconciliation oracle) and, on small grids, by an exhaustive check working directly from the definition. Only reconciled values appear. Code and data: github.com/apemm/Kagey-Problems.

the natural guess that the minimum kept set has size $(n - 1)(m - 1)$, which holds for every grid with $n \leq 4$ and fails at 5×5 ; the prediction was registered before the computation was run.

Because much of the evidence here is computational, we state the verification protocol. Small cases were solved by hand and recorded as a fixed ledger before any code was written. Every reported value was then produced by two independently written implementations, an ILP formulation over the pruned difference hypergraph and a branch-and-bound search using no ILP, and on small grids by a third exhaustive check that works directly from the definition of identifiability and never forms difference sets. Only values on which the implementations agree are reported. Conjectured formulas were recorded, with their predictions for untested grids, before those grids were computed; the refutation above is one such prediction failing.

The paper is organized as follows. Section 2 fixes conventions and defines the presentation map. Section 3 proves the reduction and its consequences. Section 4 describes the pruning of the difference hypergraph. Section 5 proves the Strip Theorem. Section 6 reports the computations and the verification. Section 7 treats uniqueness at 5×5 . Section 8 records the refuted and live conjectures and states open questions.

2 Conventions and the presentation map

Vertices of the $n \times m$ grid are denoted as pairs (i, j) with $1 \leq i \leq m$ (column) and $1 \leq j \leq n$ (row). The set of all vertices on the grid is V . The rectangle class consists of all rectangles with four corners on grid points, tilted rectangles included and degenerate (zero-area) configurations excluded; $C(R)$ denotes the corner set of a rectangle R . For a blackout $S \subseteq V$ the *presentation map* sends each rectangle to its visible corners, $P_S(R) = C(R) \setminus S$, and S is *valid* when P_S is injective over all rectangles. The empty presentation is an ordinary and accepted value of the map, that at most one rectangle may be fully hidden is a corollary of injectivity, not an axiom (Remark 1).

Remark 1. Suppose 2 distinct rectangles $R_1 \neq R_2$ with corner sets C_1 and C_2 are both fully hidden, that is, $C_1 \subseteq S$ and $C_2 \subseteq S$. Then $P_S(R_1) = \emptyset = P_S(R_2)$, and since $\emptyset$ is an ordinary value of the presentation map, injectivity gives $R_1 = R_2$, contradicting our supposition. Hence, at most 1 rectangle may be fully hidden under any valid blackout, with no separate convention required.

3 The reduction

Lemma 1 (difference-set characterization). *For any sets A, B, S : $A \setminus S = B \setminus S$ if and only if $A \triangle B \subseteq S$.*

Proof. ($\Rightarrow$) Assume $A \setminus S = B \setminus S$, and let $x \in A \triangle B$ be arbitrary. Exchanging A and B makes no difference since $\triangle$ is commutative and symmetric about A and B , so we may assume $x \in A \setminus B$. Suppose $x \notin S$. Then $x \in A \setminus S$, and since $A \setminus S = B \setminus S$ we get $x \in B \setminus S$, hence $x \in B$, contradicting $x \in A \setminus B$. Therefore $x \in S$. As x was arbitrary, $A \triangle B \subseteq S$.

($\Leftarrow$) Assume $A \triangle B \subseteq S$. Both the hypothesis and the conclusion are unchanged when A and B are exchanged, so it suffices to prove $A \setminus S \subseteq B \setminus S$. Let $x \in A \setminus S$ be arbitrary, so $x \in A$ and $x \notin S$. Suppose $x \notin B$. Then $x \in A \setminus B \subseteq A \triangle B \subseteq S$, contradicting $x \notin S$. Hence $x \in B$, and with $x \notin S$ we get $x \in B \setminus S$. As x was arbitrary, $A \setminus S \subseteq B \setminus S$; the reverse inclusion follows by symmetry, so $A \setminus S = B \setminus S$. $\square$

Corollary 1. *S is valid if and only if the kept set $K = V \setminus S$ intersects the difference set $C(R_1) \triangle C(R_2)$ for every pair of distinct rectangles. Hence a maximum blackout is the complement of a minimum hitting set of the difference hypergraph.*

Proof. By definition, S is valid if and only if $P_S(R_1) \neq P_S(R_2)$, i.e. $C(R_1) \setminus S \neq C(R_2) \setminus S$, for every pair of distinct rectangles. From Lemma 1, this holds if and only if $C(R_1) \triangle C(R_2) \not\subseteq S$, which is exactly the statement that at least 1 point of the difference set lies in K . $\square$

4 Preprocessing

Prior to any computation, we prune the hypergraph for convenience and clarity. The first technique is deduplication, since identical difference sets impose identical constraints on the allowed K , so we only need to keep one copy. The second technique is a containment rule: if D_1 and D_2 are two difference sets and $D_1 \subset D_2$, any K intersecting D_1 automatically intersects D_2 as well, so D_2 can be omitted from the hypergraph without any impact on which K are valid. Third, an empty row remark: an empty difference set would mean that $C_1 \triangle C_2 = \emptyset$, hence $C_1 = C_2$, no K can intersect this difference set, so its presence would signal that the rectangle list was not deduplicated rather than a genuine constraint.

With these pruning techniques, the hypergraph can be replaced by its pruned antichain of minimal rows. All of the following computations in Section 6 run on the pruned version of the hypergraph.

5 The Strip Theorem

Theorem 1 (Strip Theorem). *For a $2 \times m$ grid with $m \geq 3$, every maximum valid blackout has size $m + 1$, there are exactly $m \cdot 2^{m-1}$ of them, and they are characterized as the complements of kept sets containing one survivor per column and missing exactly one column. For $m = 2$ the condition is vacuous and every blackout is valid.*

Proof. Define K as the visible set, where $V \setminus S = K$, and let K_0 denote the minimum possible size of K over valid blackouts. For a $2 \times m$ grid, define the set T to be the set of all columns with at least 1 of their 2 points in K . For $m = 2$ there is exactly one rectangle, so injectivity is vacuous and every blackout is valid; assume $m \geq 3$. Suppose S is a maximum valid blackout.

On a $2 \times m$ strip every rectangle is axis-aligned, since a tilted rectangle's corners would occupy at least 3 distinct rows and the strip has only 2. We can represent any possible rectangle R as $R_{i,j}$, where i is the column containing the left corners of R and j is the column containing the right corners, $1 \leq i < j \leq m$.

If two distinct rectangles share exactly one column, say $R_{z,p}$ and $R_{z,q}$, the shared column's corners overlap, and $C(R_{z,p}) \triangle C(R_{z,q})$ is exactly the 4 points of columns p and q ; we call this the pair row on $\{p, q\}$. If they share no column, the difference set is all 8 points of 4 distinct columns, which contains a pair row and is deleted in preprocessing (Section 4). Every pair $\{p, q\}$ arises this way: two rectangles sharing a column span at least 3 distinct columns between them, and $m \geq 3$ guarantees a third column z from which to build $R_{z,p}$ and $R_{z,q}$. Hence, the pruned hypergraph consists of exactly the $\binom{m}{2}$ pair rows.

Then, from Lemma 1, a valid visible set K is one that intersects every pair row. Conditioning on T , this constrains T to contain every column except at most one: if two distinct columns p and q are excluded, the pair row on $\{p, q\}$ contains no point of K , and K would not intersect every pair row. Each column of T contributes at least 1 point, so $|K| \geq |T| \geq m - 1$, and hence $K_0 = m - 1$ and $|S| \leq 2m - (m - 1) = m + 1$. If no column is fully blacked out, T would contain every column and K would need m points, so $|S| \leq m$. If any column of T kept both of its points, $|K| \geq m$ as well. Hence, at the maximum, exactly one column is excluded from T , and in the other $m - 1$ columns, exactly 1 of the 2 points in each column is blacked out.

Every such construction is valid: any pair of columns includes at least one column of T , whose surviving point meets the pair row. There are m ways to choose which column to exclude from T and 2^{m-1} ways to choose which 1 of the 2 points to black out in the remaining $m - 1$ columns. Hence, $m \cdot 2^{m-1}$ such distinct maximum blackouts exist, characterized as in the statement. $\square$

6 Computations and verification

Every value below was produced by the author’s engine (ILP over the pruned difference hypergraph) and reconciled against an independently written oracle (branch and bound, no ILP) and, on small grids, an exhaustive definitional check that never forms difference sets. Rectangle counts match the literature: $n \times n$ counts 1, 10, 44, 130 agree with OEIS A085582, and $3 \times m$ counts with OEIS A289832 (indexing note in the bibliography).

grid	rects	blackout	kept	optima	verification
2×2	1	4	0	1	three-way
2×3	3	4	2	12	three-way
2×4	6	5	3	32	three-way
2×5	10	6	4	80	three-way; pre-registered
2×6	15	7	5	192	engine + oracle
2×7	21	8	6	448	engine + oracle
3×3	10	5	4	53	three-way
3×4	20	6	6	224	three-way
3×5	33	7	8	892	three-way
3×6	49	8	10	3420	three-way
4×4	44	7	9	316	three-way
4×5	74	8	12	1810	three-way
3×7	68	9	12	12704	engine + brute
3×8	90	10	14	45864	engine + brute
3×9	115	11	16	161992	engine + brute
3×10	143	12	18	—	engine only (value)
4×6	110	10	14	26	engine + brute
4×7	152	11	17	292	engine + brute
4×8	200	12	20	—	engine only (value)
5×5	130	10	15	8	engine + brute
5×6	198	11	19	—	engine only (value)
5×7	276	12	23	—	engine only (value)
6×6	313	12	24	—	engine only (value)

Table 1: Exact values. “Three-way” = engine, oracle, and definitional brute agree; “engine + brute” = the ILP engine and the exhaustive definitional check agree; “engine only” = the exhaustive check is infeasible at that size.

7 Uniqueness at 5×5

Theorem 2 (computational). *The 5×5 grid has maximum blackout 10, attained by exactly 8 blackouts, and the maximum blackout is unique up to the symmetries of the square: the 8 optima*

form a single orbit of the dihedral group D_4 , with trivial stabilizer.

The value and the count were produced independently by the engine and by the exhaustive definitional check; the orbit claim was verified by applying the 8 symmetries of the grid to one optimum and recovering all 8. The unique extremal blackout is the 9-point anti-diagonal staircase, the old $n + m - 1$ pattern, plus 1 detached corner: the +1 made visible. A representative (‘#’ blacked out, ‘.’ kept):

```
..###
..#..
.##..
##...
#...#
```

The kept set is two triangular blocks of 5 and 10 points on either side of the band, and the extremal object is unique up to symmetry while having no symmetry of its own.

8 Refuted and live conjectures

conjecture	status	killed by
kept = $(n - 1)(m - 1)$	refuted (pre-registered)	5×5
H_{sum} : blackout = $n + m - 1 + \lfloor n + m \geq 10 \rfloor$	refuted (pre-registered)	3×7
H_{area} : blackout = $n + m - 1 + \lfloor nm \geq 24 \rfloor$	refuted (pre-registered)	3×8
Four-Row: kept($4, m$) = $\lceil 5m/2 \rceil - 1$	refuted (pre-registered)	4×8

Conjecture 1 (Three-Row). blackout($3, m$) = $m + 2$ for all $m \geq 3$.

Verified for $3 \leq m \leq 10$, eight cells, every prediction registered before its computation ran. This is the shape of the Strip Theorem one row up, kept set $2(m - 1)$ against the strip’s $m - 1$, and the strip argument is the plausible route; the difference sets on three rows are no longer full column pairs, so the classification step is where new work begins.

Conjecture 2 (Excess-1). blackout(n, m) = $n + m$ for $\min(n, m) \geq 4$, *excepting* 4×4 and 4×5 , where it is $n + m - 1$.

Verified on seven cells (4×6 through 6×6), including the 6×6 prediction of 12, registered before the 1063-second computation confirmed it. The break past $n + m - 1$ is a min-dimension phenomenon: every three-row grid obeys $m + 2$, while every cell with $\min(n, m) \geq 4$ outside the two exceptions exceeds $n + m - 1$ by exactly 1.

Beyond the two conjectures, three questions strike us as the natural next ones: why the number of optima collapses at 4×6 (26, against neighbors in the hundreds and thousands); whether the excess ever grows past 1, with 5×8 and 6×7 registered at 13 as the next kill attempts; and whether the 5×5 uniqueness admits a structural proof rather than a computational one.

Data and code availability

The engine, the reconciliation oracle, the definitional brute, the pruned tables, and the ledger of pre-registrations and refutations accompany this note at <https://github.com/apemm/Kagey-Problems>.

References

- [1] P. Kagey, Problem 001, Open Problem Collection, peterkagey.com/problems/001/ (accessed August 2026).
- [2] OEIS Foundation, Sequence A085582: number of rectangles (orthogonal or not) with corners on an $n \times n$ grid of points. <https://oeis.org/A085582>. Verified against Table 1: 1, 10, 44, 130 at $n = 2..5$.
- [3] OEIS Foundation, Sequence A289832: table of rectangle counts on rectangular grids; diagonal equals A085582. <https://oeis.org/A289832>. Indexing: $T(n, k)$ counts rectangles on an $(n+1) \times (k+1)$ point grid, so an $r \times c$ point grid appears as $T(c-1, r-1)$; e.g. our 3×7 count is $T(6, 2) = 68$.